

Signal	Direction	Behavior	Manual	GPT-o4-mini	Sonnet-4	GPT-5.1
M0 Macrophage						
Pro-Inflammatory Factor	Increases	Transform To M1 Macrophage				
Anti-Inflammatory Factor	Increases	Transform To M2 Macrophage				
Necrotic Debris	Increases	Transform To M1 Macrophage				
Apoptotic Debris	Decreases	Migration Speed				
Necrotic Debris	Decreases	Migration Speed				
Volume	Decreases	Phagocytose Apoptotic Cell				
Volume	Decreases	Phagocytose Necrotic Cell				
Substrate Gradient: Apoptotic Debris	Increases	Chemotactic Response To Substrate				
M1 Macrophage						
Pro-Inflammatory Factor	Increases	Chemotactic Response To Pro-Inflammatory...				
Oxygen	Decreases	Transform To M2 Macrophage				
Apoptotic Debris	Decreases	Migration Speed				
Necrotic Debris	Decreases	Migration Speed				
Volume	Decreases	Phagocytose Apoptotic Cell				
Volume	Decreases	Phagocytose Necrotic Cell				
Substrate Gradient: Necrotic Debris	Increases	Phagocytose Necrotic Cell				
Contact With Malignant Epithelial Cell	Increases	Attack Malignant Epithelial Cell				
Apoptotic Debris	Increases	Apoptotic Debris Uptake				
Pro-Inflammatory Factor	Increases	Attack Malignant Epithelial Cell				
M2 Macrophage						
Apoptotic Debris	Decreases	Migration Speed				
Necrotic Debris	Decreases	Migration Speed				
Volume	Decreases	Phagocytose Apoptotic Cell				
Volume	Decreases	Phagocytose Necrotic Cell				
Substrate: Anti-Inflammatory Factor	Increases	Migration Speed				
Contact With Malignant Epithelial Cell	Decreases	Migration Speed				
Apoptotic Debris	Increases	Apoptotic Debris Uptake				
Anti-Inflammatory Factor	Increases	Chemotactic Response To Anti-Inflammatory...				
Anti-Inflammatory Factor	Increases	Phagocytose Apoptotic Cell				
Effector T Cell						
Anti-Inflammatory Factor	Increases	Transform To Exhausted T Cell				
Pro-Inflammatory Factor	Increases	Attack Malignant Epithelial Cell				
Pro-Inflammatory Factor	Increases	Migration Speed				
Pro-Inflammatory Factor	Increases	Chemotactic Response To Pro-Inflammatory...				
Contact With Malignant Epithelial Cell	Decreases	Migration Speed				
Substrate: Pro-Inflammatory Factor	Increases	Attack Cell Type Malignant Epithelial Cell				
Contact With Malignant Epithelial Cell	Increases	Attack Malignant Epithelial Cell				
Oxygen	Increases	Attack Malignant Epithelial Cell				
Exhausted T Cell						
Anti-Inflammatory Factor	Decreases	Migration Speed				
Contact With Malignant Epithelial Cell	Decreases	Migration Speed				
Anti-Inflammatory Factor	Decreases	Attack Malignant Epithelial Cell				
Malignant Epithelial Cell						
Oxygen	Decreases	Necrosis				
Oxygen	Increases	Cycle Entry				
Oxygen	Decreases	Migration Speed				
Pressure	Decreases	Cycle Entry				
Apoptotic	Increases	Apoptotic Debris Secretion				
Necrotic	Increases	Necrotic Debris Secretion				
Damage	Increases	Apoptosis				
Substrate: Oxygen	Increases	Cycle Entry				
Contact With Effector T Cell	Increases	Pro-Inflammatory Factor Secretion				
Oxygen	Decreases	Cycle Entry				